

A Sharp M-Matrix Threshold for a Two-Dimensional Oblique Skorokhod Map

Route-A source-local certificate HCS-C346

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Abstract

For oblique regulation in the quadrant with the stated two-axis reflection matrix R , we prove the *fixed point and sharp M-matrix chamber*: every càdlàg input has one and only one regulated path precisely when $\rho\sigma < 1$. We add *weighted stability, Picard, and time change*: the natural norm has contraction factor $\sqrt{\rho\sigma}$, explicit solution stability, monotone geometric iteration, causality and reparameterization covariance. Finally, *sharp wall, executable audit, and Route-A boundary* close critical nonuniqueness, critical/supercritical nonexistence, triangular and post-jump faces without upgrading finite evidence to an all-path proof.

1 Convention and the sharp existence theorem

Fix $T < \infty$, $\rho, \sigma \geq 0$, and

$$R = \begin{pmatrix} 1 & -\rho \\ -\sigma & 1 \end{pmatrix}.$$

Let $x = (x_1, x_2) \in D([0, T], \mathbb{R}^2)$ with $x(0) \geq 0$ coordinatewise. A solution consists of càdlàg z, y such that

$$z = x + Ry \geq 0, \quad y(0) = 0, \quad y_i \text{ is nondecreasing}, \quad \int_{[0, t]} z_i(s) dy_i(s) = 0. \quad (1)$$

The integrand is the *post-jump* state. Thus a regulator jump is supported on an axis after the jump, including simultaneous corner jumps.

For scalar càdlàg f , write

$$(\mathcal{L}f)(t) = \sup_{0 \leq s \leq t} [-f(s)]_+.$$

Lemma 1 (One-sided regulator). *If $f(0) \geq 0$, the unique nondecreasing càdlàg r , with $r(0) = 0$, for which $g = f + r \geq 0$ and $\int g dr = 0$ is $r = \mathcal{L}f$. Moreover,*

$$\|\mathcal{L}f - \mathcal{L}h\|_\infty \leq \|f - h\|_\infty. \quad (2)$$

Proof. The running supremum is the least nondecreasing correction making $f + r$ nonnegative. It increases only at a new negative running minimum, where the post-jump corrected state is zero. Conversely, feasibility forces every regulator to dominate $\mathcal{L}f$. A first strict excess would be an increase with positive corrected state, contradicting complementarity. Finally, positive part and running supremum are one-Lipschitz, proving (2). $\square$

Theorem 2 (Sharp all-input threshold). *Problem (1) has exactly one solution for every admissible input and every finite horizon if and only if*

$$\rho\sigma < 1. \quad (3)$$

Inside this chamber, its regulator is the unique solution of

$$y_1 = \mathcal{L}(x_1 - \rho y_2), \quad y_2 = \mathcal{L}(x_2 - \sigma y_1). \quad (4)$$

Proof. With the other coordinate held fixed, each line of (1) is the scalar problem in Lemma 1; hence (1) and (4) are equivalent. Conversely, at time zero the fixed-point equations and $x(0) \geq 0$ imply $y_1(0) \leq \rho y_2(0)$ and $y_2(0) \leq \sigma y_1(0)$; since $\rho\sigma < 1$, both values vanish. Thus the fixed point has the required initial condition.

First suppose $\rho, \sigma > 0$, put $w_1 = \sqrt{\rho}$, $w_2 = \sqrt{\sigma}$, $q = \sqrt{\rho\sigma}$, and define

$$\|u\|_w = \max_{i=1,2} \frac{\|u_i\|_\infty}{w_i}.$$

For the right side Φ_x of (4), inequality (2) gives

$$\|\Phi_x(u) - \Phi_x(v)\|_w \leq q\|u - v\|_w.$$

Thus (3) makes Φ_x a contraction on the complete space of bounded càdlàg pairs. Its unique fixed point is càdlàg and solves the problem. The constant q is the best uniform coefficient: each weighted cross-coordinate estimate can be attained by a constant difference after a suitable downward input shift.

If $\rho = 0$, solve successively $y_1 = \mathcal{L}x_1$ and $y_2 = \mathcal{L}(x_2 - \sigma y_1)$; if $\sigma = 0$, reverse the order. This also covers $\rho = \sigma = 0$ and proves sufficiency.

For necessity, let $\rho\sigma = 1$. Then $R(\rho, 1)^\top = 0$, so zero input has the distinct solutions

$$y = (\rho h, h), \quad z = 0, \quad (5)$$

for every nondecreasing càdlàg h with $h(0) = 0$. More strongly, whenever $\rho\sigma \geq 1$, an input jumping from zero to $(-1, -1)$ cannot be feasible: the post-jump values would obey

$$y_1 \geq 1 + \rho y_2, \quad y_2 \geq 1 + \sigma y_1,$$

and hence $(1 - \rho\sigma)y_1 \geq 1 + \rho$, an impossibility. Therefore no point on or beyond the wall is well posed for every input. $\square$

2 Quantitative stability and path structure

Theorem 3 (Stability and constructive convergence). *Assume $\rho, \sigma > 0$ and $q < 1$. If inputs x, x' have solutions $(z, y), (z', y')$, then*

$$\|y - y'\|_w \leq \frac{\|x - x'\|_w}{1 - q}, \quad \|z - z'\|_w \leq \frac{2\|x - x'\|_w}{1 - q}. \quad (6)$$

Starting with $y^{(0)} = 0$ and setting $y^{(n+1)} = \Phi_x(y^{(n)})$ gives a coordinatewise increasing sequence and

$$\|y^{(n+1)} - y^{(n)}\|_w \leq q^n \|y^{(1)}\|_w, \quad (7)$$

$$\|y - y^{(n)}\|_w \leq \frac{q^n}{1 - q} \|y^{(1)}\|_w. \quad (8)$$

The solution map is causal, preserves continuity, and commutes with continuous nondecreasing onto time changes that preserve the endpoints.

Proof. For two inputs, the scalar estimate gives

$$\|\Phi_x(u) - \Phi_{x'}(v)\|_w \leq \|x - x'\|_w + q\|u - v\|_w.$$

Evaluate it at the two fixed points for the first inequality in (6). Each normalized row of R has weighted absolute sum $1 + q$, whence

$$\|R(y - y')\|_w \leq (1 + q)\|y - y'\|_w.$$

Using $z - z' = x - x' + R(y - y')$ yields the second inequality. The map Φ_x is order preserving and $0 \leq \Phi_x(0)$, so Picard iterates increase. Contraction gives (7), and summing its geometric tail gives (8).

Running suprema use only the past, so uniqueness gives causality. They map continuous functions to continuous functions; uniform Picard convergence then preserves continuity. If $\lambda : [0, T'] \rightarrow [0, T]$ is continuous, nondecreasing, onto, and endpoint preserving, its initial images satisfy $\lambda([0, t]) = [0, \lambda(t)]$. Consequently

$$\mathcal{L}(f \circ \lambda) = (\mathcal{L}f) \circ \lambda.$$

Compose (4) with λ and use uniqueness. Flat pieces of λ simply pause both state and regulator. $\square$

Remark 4. *When one weight vanishes, the weighted norm is not defined. The triangular formulas in Theorem 2 are the correct boundary statement; no singular limiting norm is smuggled into that face.*

3 Boundary closure and executable receipt

At the critical wall, equation (5) proves nonuniqueness even for the smoothest possible input, whereas the negative jump proves nonexistence. These are separate failures. At normal reflection the solution is $(\mathcal{L}x_1, \mathcal{L}x_2)$; a one-sided coupling is triangular. The post-jump convention resolves a step input through the linear complementarity problem for $d = y - y^-$ and the new state $x + R(y^- + d)$. Strict inequality in (3), the off-diagonal signs, and simultaneous cascades are all therefore essential.

The machine-readable receipt has six rational matrices, 36 original events, 72 pause-expanded events, 27 Picard rows, three critical nonunique regulators, three no-solution witnesses, and 693 audited scalar leaves. An independent implementation performs 886 checks; a separate SymPy lane closes 5,125 exact identities; two isolated reproductions are byte-identical; and 68 parser, semantic, stale-hash and repaired-hash attacks are rejected. These checks audit conventions. Theorems 2–3, not finite enumeration, prove the all-input result.

The nearest workspace owners are scalar Moreau play (C332), one-interface skew Brownian motion (C266), total-variation flow (C279), and dry-friction capture (C238). None owns a two-axis oblique reflection map or its sharp M-matrix threshold.

The strict evaluation is

$$(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL).$$

Free couplings and driven contacts have no rational-prime taxonomy; $\det R$ is not an Euler product or target determinant; and the dissipative regulator provides no natural target-zero quantization. Route A is rejected and Route B is locked under NO_BAD_EULER_OR_ROOT_NUMBER. No target arithmetic local data, Euler factor, root number, automorphy, divisor or counting law, functional equation, zero match, or Hilbert–Pólya operator is claimed.

AI use. A generative language model assisted drafting and code scaffolding. The self-contained proof, independent reconstruction, hostile mutations and deterministic artifacts define the audit record.

Source lineage

This source-local reconstruction makes no literature-priority claim.

References

- [1] J. M. Harrison and M. I. Reiman, “Reflected Brownian Motion on an Orthant,” *Annals of Probability* 9 (1981), 302–308. DOI: [10.1214/aop/1176994471](https://doi.org/10.1214/aop/1176994471).
- [2] P. Dupuis and H. Ishii, “On Lipschitz Continuity of the Solution Mapping to the Skorokhod Problem, with Applications,” *Stochastics and Stochastics Reports* 35 (1991), 31–62. DOI: [10.1080/17442509108833688](https://doi.org/10.1080/17442509108833688).